

Curriculum Vitae

Experience and Projects

FRTB Analyst Intern at HSBC: Worked on the Stress scenario risk management dashboard for the FRTB team. Built plots and pivot tables to analyse the full reval and sensi based SSRM of a portfolio of NMRF(non modellable risk factors) including volatility smiles and forward dividends in equities, FX and commodities. Also worked on capital allocation of ES risk measure across various risk types.

Teaching Assistant: Grader for Fixed Income course, part of the Masters In Computational Finance program, covering interest rate securities

Bond Portfolio Allocation Under Interest Rate Risk: Formulated an optimal treasury allocation problem using MDPs motivated by the collapse of Silicon Valley Bank. Minimized 95th percentile of portfolio losses by 10%.

Hedging Using Options: Simulated stock price paths and applied stochastic-volatility models (Dupire, Heston, Breeden-Litzenberger's approximation) to a CVaR-based optimal portfolio framework on NIFTY 50 options. Analyzed volatility skew and smile behaviour.

Summer Intern at Stochastic Systems Lab, IEOR Purdue University: Worked on RL algorithms for stochastic impulse control problems, using neural networks to solve PDEs arising from the Feynman-Kac equation. Also explored estimation of parameters using score-based techniques.

Education and Achievements

Carnegie Mellon University	Pittsburgh
<i>PhD student in Applied Mathematics</i>	<i>2026–2031</i>

Indian Institute of Science (IISc)	Bangalore
<i>B.Tech in Mathematics and Computing, CGPA: 9.2/10.0</i>	<i>2022–2026</i>
Entrance Exam: JEE Advanced (All India Rank 225)	

Key Courses: Probability Theory, Stochastic Calculus, Time Series, Random Processes, Numerical Methods, Reinforcement Learning, Applied DS/AI, Scalable Systems, Finance and Risk Management.

Key UG Course Projects:

- Implemented core RL algorithms (Deterministic Actor–Critic, Q-learning, TRPO, TD-learning), ARIMA/ARMA time-series models, AIC-based model selection, mean-variance portfolio optimization, Longstaff–Schwartz for American and Bermudan options, Black–Scholes pricing, and bootstrapped ZCYC from historical G-Sec data.
- Built a market-research automation app using Firebase, LangChain, and BeautifulSoup.
- Presented a term paper on diffusion models and used HPC/PySpark for large-scale data processing.

Skills and Extracurricular Activities

- **Quant:** Stochastic Calculus, Derivatives, Numerical PDEs, Monte Carlo, Statistical Methods
- **Languages:** Python, C++
- **Founding Member, Quant Finance Club:** Gave a talk on the Black–Scholes equation [Talks](#)